Diffraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain spreading around edges and through openings.

SUCCESS CRITERIA - I can...

1. Compare opening size with wavelength.
2. Distinguish diffraction from refraction.
3. Use a wave picture to justify spreading.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “diffraction” and “wavelength” to explain your model or result. Sentence starters are optional.

diffraction: wave spreading around an edge or through an opening

wavelength: distance between corresponding points on repeated waves

boundary: the place where a medium or constraint changes

01 Bring back an earlier idea

A wave has frequency 4 Hz and wavelength 6 m.
Find its speed.

Word help: frequency = number of cycles each second

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: λ = v/f; compare opening width with λ. Spreading is pronounced when the opening width is comparable to wavelength. Label the opening and wavefronts.

Worked reasoning: A wave with wavelength 1 m meets openings 1 m and 10 m wide. Spreading relative to opening size is more noticeable for the 1 m opening.

Your turn: Draw incoming parallel wavefronts and the emerging patterns for each opening.

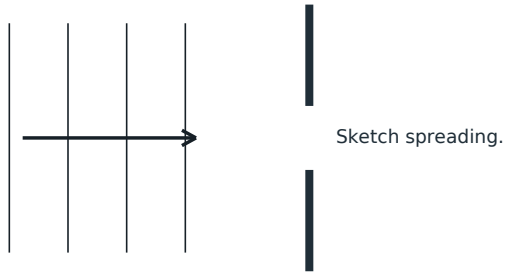
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

Sketch wavefronts after a narrow opening about one wavelength wide.
Compare with a much wider opening.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

04 Build the solution

Opening A is 1 m wide, opening B is 10 m; wavelength is 1 m. Which shows more noticeable spreading relative to opening width?
Explain.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A student calls all wave bending “refraction.” Explain how diffraction can happen without crossing into a different-speed medium.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

At the same kelvin temperature, sample B contains 4 times as many particles of the same ideal gas as A. **Compare** average energy per particle and total translational energy.

Word help: temperature = a measure of thermal state; it is not an energy amount

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

For the same opening, would a longer wavelength generally spread more or less?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Keep opening width fixed and halve wavelength. **Predict** the change in relative spreading and draw before/after patterns.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3